

# Incompressible Isotropic Hyperelasticity: Constitutive Models and Homogeneous Solutions

This note collects the standard incompressible isotropic hyperelastic models and the closed-form stress states they predict under homogeneous deformation (uniaxial, equibiaxial, pure shear, simple shear). The homogeneous solutions serve as reference solutions for the mixed displacement–pressure formulation of the framework (**formulation: mixed** with any of the six models at  $\kappa = \infty$ ; Section 5.5). Conventions follow the code: the pressure unknown  $p$  is the tensile-positive mean stress, and in two dimensions the kernels are plane strain with  $F_{33} = 1$  unless plane stress is selected (Section 5.3). The weak forms themselves are in `solid_mechanics_forms.tex`.

## 1 Kinematics and the Incompressibility Constraint

Let  $\Omega_R$  be the reference configuration; referential quantities carry the subscript  $R$ . With  $\mathbf{x} = \boldsymbol{\varphi}(\mathbf{X}) = \mathbf{X} + \mathbf{u}(\mathbf{X})$ ,  $\mathbf{X} \in \Omega_R$ , the deformation gradient, Jacobian, right and left Cauchy–Green tensors are

$$\mathbf{F} = \mathbf{I} + \text{Grad } \mathbf{u}, \quad J = \det \mathbf{F}, \quad \mathbf{C} = \mathbf{F}^T \mathbf{F}, \quad \mathbf{B} = \mathbf{F} \mathbf{F}^T. \quad (1)$$

The principal invariants of  $\mathbf{C}$  (equivalently of  $\mathbf{B}$ ) are

$$I_1 = \text{tr } \mathbf{C}, \quad I_2 = \frac{1}{2}[(\text{tr } \mathbf{C})^2 - \text{tr}(\mathbf{C}^2)] = \text{tr}(\text{cof } \mathbf{C}), \quad I_3 = \det \mathbf{C} = J^2. \quad (2)$$

In terms of the principal stretches  $\lambda_i$  (the square roots of the eigenvalues of  $\mathbf{C}$  and  $\mathbf{B}$ ),

$$I_1 = \lambda_1^2 + \lambda_2^2 + \lambda_3^2, \quad I_2 = \lambda_1^2 \lambda_2^2 + \lambda_2^2 \lambda_3^2 + \lambda_3^2 \lambda_1^2, \quad J = \lambda_1 \lambda_2 \lambda_3. \quad (3)$$

An incompressible material satisfies the internal constraint

$$J = 1 \quad (\lambda_1 \lambda_2 \lambda_3 = 1), \quad (4)$$

under which two identities used repeatedly below hold:

$$I_2 = \lambda_1^{-2} + \lambda_2^{-2} + \lambda_3^{-2} = \text{tr } \mathbf{B}^{-1}, \quad \mathbf{B}^2 = I_1 \mathbf{B} - I_2 \mathbf{I} + \mathbf{B}^{-1} \quad (\text{Cayley–Hamilton with } I_3 = 1). \quad (5)$$

**Isochoric split.** The framework writes its decoupled models in terms of the isochoric deformation gradient and modified invariants

$$\bar{\mathbf{F}} = J^{-1/3} \mathbf{F}, \quad \bar{I}_1 = J^{-2/3} I_1, \quad \bar{I}_2 = J^{-4/3} I_2, \quad (6)$$

so that the same energy makes sense at finite bulk modulus. On an incompressible solution  $J = 1$  and the barred and unbarred quantities coincide; the formulas below are written with  $I_1, I_2$ .

## 2 Constitutive Framework

### 2.1 Strain energy and the reaction stress

For an isotropic incompressible material the strain energy per unit reference volume is a function of the two remaining invariants,  $\Psi = \Psi(I_1, I_2)$ , or equivalently a symmetric function of the principal stretches subject to (4). Write

$$\Psi_1 = \frac{\partial \Psi}{\partial I_1}, \quad \Psi_2 = \frac{\partial \Psi}{\partial I_2}. \quad (7)$$

The constraint is enforced by a Lagrange multiplier  $q$  that does no work, and the Cauchy stress is

$$\boldsymbol{\sigma} = 2\Psi_1 \mathbf{B} + 2\Psi_2 (I_1 \mathbf{B} - \mathbf{B}^2) - q \mathbf{I} = 2\Psi_1 \mathbf{B} - 2\Psi_2 \mathbf{B}^{-1} + (2\Psi_2 I_2 - q) \mathbf{I}, \quad (8)$$

where the second form uses (5). The multiplier  $q$  is not given by the constitutive law; it is determined by equilibrium and the boundary conditions. Only the isotropic part of  $\boldsymbol{\sigma}$  is affected, so the convention-free statement of (8) is the deviatoric–mean split

$$\boxed{\boldsymbol{\sigma} = \boldsymbol{\sigma}' + p \mathbf{I}, \quad \boldsymbol{\sigma}' = 2\Psi_1 \operatorname{dev} \mathbf{B} - 2\Psi_2 \operatorname{dev} \mathbf{B}^{-1}, \quad p = \frac{1}{3} \operatorname{tr} \boldsymbol{\sigma},} \quad (9)$$

with  $\operatorname{tr} \mathbf{B} = I_1$  and  $\operatorname{tr} \mathbf{B}^{-1} = I_2$  in the deviators.

**Sign convention.** Much of the literature writes  $\boldsymbol{\sigma} = -\tilde{p} \mathbf{I} + 2\Psi_1 \mathbf{B} - 2\Psi_2 \mathbf{B}^{-1}$  and calls  $\tilde{p}$  the hydrostatic pressure. Because different authors absorb different isotropic terms into  $\tilde{p}$ , its value is convention-dependent;  $p = \frac{1}{3} \operatorname{tr} \boldsymbol{\sigma}$  (tensile positive) is not, and it is the quantity the mixed formulation solves for (Section 2.5). All results below are stated in terms of  $p$ .

### 2.2 Principal-stretch form

In the principal frame  $\mathbf{B} = \sum_i \lambda_i^2 \mathbf{n}_i \otimes \mathbf{n}_i$  and  $\boldsymbol{\sigma} = \sum_i \sigma_i \mathbf{n}_i \otimes \mathbf{n}_i$  with

$$\sigma_i = 2\Psi_1 \lambda_i^2 - 2\Psi_2 \lambda_i^{-2} - q = \lambda_i \frac{\partial \Psi}{\partial \lambda_i} - q', \quad (10)$$

where  $q' = q - 2\Psi_2 I_2$  and the second form follows from  $\partial I_1 / \partial \lambda_i = 2\lambda_i$ ,  $\partial I_2 / \partial \lambda_i = 2\lambda_i(I_1 - \lambda_i^2)$  together with (4):

$$\lambda_i \frac{\partial \Psi}{\partial \lambda_i} = 2\Psi_1 \lambda_i^2 - 2\Psi_2 \lambda_i^{-2} + 2\Psi_2 I_2. \quad (11)$$

The principal stress differences are constitutive,

$$\sigma_i - \sigma_j = 2\Psi_1(\lambda_i^2 - \lambda_j^2) - 2\Psi_2(\lambda_i^{-2} - \lambda_j^{-2}) = \lambda_i \frac{\partial \Psi}{\partial \lambda_i} - \lambda_j \frac{\partial \Psi}{\partial \lambda_j}, \quad (12)$$

and the recipe for every homogeneous problem is the same: the two differences follow from (12), one principal stress is fixed by a traction-free face, and then  $p = \frac{1}{3}(\sigma_1 + \sigma_2 + \sigma_3)$ .

### 2.3 Other stress measures

With  $J = 1$  the first and second Piola–Kirchhoff stresses are

$$\mathbf{P} = \boldsymbol{\sigma} \mathbf{F}^{-T}, \quad \mathbf{S} = \mathbf{F}^{-1} \boldsymbol{\sigma} \mathbf{F}^{-T}, \quad (13)$$

and for a diagonal  $\mathbf{F} = \text{diag}(\lambda_1, \lambda_2, \lambda_3)$

$$P_i = \frac{\sigma_i}{\lambda_i}, \quad S_i = \frac{\sigma_i}{\lambda_i^2}. \quad (14)$$

$P_i$  is the nominal stress (force per unit reference area), which is what a traction condition of the total Lagrangian formulation prescribes:  $\mathbf{t}_R = \mathbf{P}\mathbf{N}$  on the reference boundary  $\partial\Omega_R$ .

## 2.4 Small-strain limit

For  $\mathbf{u} = \varepsilon$ -small,  $\mathbf{B} \approx \mathbf{I} + 2\varepsilon$ ,  $\mathbf{B}^{-1} \approx \mathbf{I} - 2\varepsilon$  and  $\text{tr } \varepsilon = 0$ , so (9) gives

$$\boldsymbol{\sigma} \approx 2\mu \varepsilon + p \mathbf{I}, \quad \mu = 2(\Psi_1 + \Psi_2)|_{I_1=I_2=3}, \quad (15)$$

which identifies the small-strain shear modulus of every model. The corresponding Young's modulus and Poisson ratio are  $E = 3\mu$ ,  $\nu = 1/2$ .

## 2.5 Relation to the mixed formulation

The framework's decoupled materials provide  $\Psi_{\text{iso}}(\bar{\mathbf{F}})$  and  $\mathbf{P}_{\text{iso}} = \partial\Psi_{\text{iso}}/\partial\mathbf{F}$ . Since  $\Psi_{\text{iso}}$  is invariant under  $\mathbf{F} \rightarrow \alpha\mathbf{F}$ , Euler's theorem gives  $\mathbf{P}_{\text{iso}} : \mathbf{F} = 0$ , i.e.  $\text{tr}(J^{-1}\mathbf{P}_{\text{iso}}\mathbf{F}^T) = 0$ : the isochoric Cauchy stress is deviatoric for every  $\mathbf{F}$ , not only at  $J = 1$ . The mixed kernel uses

$$\mathbf{P} = \mathbf{P}_{\text{iso}}(\mathbf{F}) + p J \mathbf{F}^{-T} \implies \boldsymbol{\sigma} = J^{-1} \mathbf{P} \mathbf{F}^T = \underbrace{J^{-1} \mathbf{P}_{\text{iso}} \mathbf{F}^T}_{\text{deviatoric}} + p \mathbf{I}, \quad (16)$$

so its pressure unknown is exactly  $p = \frac{1}{3} \text{tr } \boldsymbol{\sigma}$  of (9), and on an incompressible solution  $J^{-1} \mathbf{P}_{\text{iso}} \mathbf{F}^T = \boldsymbol{\sigma}'$ . The constraint equation is  $\int q (J - 1 - p/\kappa) dV = 0$ : for  $\kappa = \infty$  it is (4) in weak form and  $p$  is a Lagrange multiplier; for finite  $\kappa$  it is the penalty relation  $p = \kappa(J - 1)$  and the solutions below are recovered as  $\kappa \rightarrow \infty$ . (Homogeneous solutions at finite  $\kappa$  require solving a scalar nonlinear equation for the lateral stretch and are not tabulated here.)

# 3 Constitutive Models

Each model is given by  $\Psi$ , its derivatives  $\Psi_1, \Psi_2$  (or  $\lambda_i \partial\Psi/\partial\lambda_i$ ), and its small-strain shear modulus from (15). All are normalised so that  $\Psi = 0$  and  $\boldsymbol{\sigma} = p \mathbf{I}$  in the reference configuration.

## 3.1 Neo-Hookean

$$\Psi = \frac{\mu}{2}(I_1 - 3), \quad \Psi_1 = \frac{\mu}{2}, \quad \Psi_2 = 0, \quad \boldsymbol{\sigma} = \mu \text{dev } \mathbf{B} + p \mathbf{I}. \quad (17)$$

One parameter, the shear modulus  $\mu$ . This is the Gaussian-chain model of rubber elasticity (Treloar) and the incompressible limit of `iso_neo_hookean`.

## 3.2 Mooney–Rivlin

$$\Psi = c_1(I_1 - 3) + c_2(I_2 - 3), \quad \Psi_1 = c_1, \quad \Psi_2 = c_2, \quad \mu = 2(c_1 + c_2), \quad (18)$$

$$\boldsymbol{\sigma} = 2c_1 \text{dev } \mathbf{B} - 2c_2 \text{dev } \mathbf{B}^{-1} + p \mathbf{I}. \quad (19)$$

The  $c_2$  term is what distinguishes the model from neo-Hookean: it changes the uniaxial and equibiaxial curves and the stresses in the constrained directions, whereas the pure-shear stress  $\sigma_1$  and the simple-shear stress  $\sigma_{12}$  depend on  $\mu$  alone (Section 4). This is `mooney_rivlin` in the code. The general Rivlin polynomial

$$\Psi = \sum_{i+j \geq 1} C_{ij} (I_1 - 3)^i (I_2 - 3)^j, \quad \mu = 2(C_{10} + C_{01}), \quad (20)$$

contains neo-Hookean ( $C_{10} = \mu/2$ ), Mooney–Rivlin ( $C_{10} = c_1$ ,  $C_{01} = c_2$ ) and Yeoh as special cases.

### 3.3 Yeoh

$$\Psi = C_{10}(I_1 - 3) + C_{20}(I_1 - 3)^2 + C_{30}(I_1 - 3)^3, \quad \Psi_1 = C_{10} + 2C_{20}(I_1 - 3) + 3C_{30}(I_1 - 3)^2, \quad \Psi_2 = 0, \quad (21)$$

with  $\mu = 2C_{10}$ . A cubic in  $I_1$  alone; typically  $C_{20} < 0$  and  $C_{30} > 0$ , which reproduces the softening-then-stiffening uniaxial response of filled rubbers. This is `yeoh` in the code.

### 3.4 Gent

$$\Psi = -\frac{\mu J_m}{2} \ln\left(1 - \frac{I_1 - 3}{J_m}\right), \quad \Psi_1 = \frac{\mu}{2} \frac{J_m}{J_m - (I_1 - 3)}, \quad \Psi_2 = 0, \quad I_1 - 3 < J_m. \quad (22)$$

Two parameters,  $\mu$  and the limiting value  $J_m$  of  $I_1 - 3$  (limited chain extensibility).  $J_m \rightarrow \infty$  recovers neo-Hookean. This is `gent` in the code.

### 3.5 Arruda–Boyce (eight-chain)

In the usual five-term series form,

$$\Psi = \mu \sum_{i=1}^5 \frac{C_i}{N^{i-1}} (I_1^i - 3^i), \quad C_1 = \frac{1}{2}, \quad C_2 = \frac{1}{20}, \quad C_3 = \frac{11}{1050}, \quad C_4 = \frac{19}{7000}, \quad C_5 = \frac{519}{673750}, \quad (23)$$

$$\Psi_1 = \mu \sum_{i=1}^5 \frac{i C_i}{N^{i-1}} I_1^{i-1} = \mu \left[ \frac{1}{2} + \frac{I_1}{10N} + \frac{11I_1^2}{350N^2} + \frac{19I_1^3}{1750N^3} + \frac{519I_1^4}{134750N^4} \right], \quad \Psi_2 = 0. \quad (24)$$

Here  $\mu$  is the chain-density parameter  $nkT$  and  $N$  the number of rigid links per chain (the locking stretch is  $\lambda_L = \sqrt{N}$ ). The small-strain shear modulus is not  $\mu$  but

$$\mu_0 = 2\Psi_1|_{I_1=3} = \mu \left( 1 + \frac{3}{5N} + \frac{99}{175N^2} + \frac{513}{875N^3} + \frac{42039}{67375N^4} \right). \quad (25)$$

The series form is `arruda_boyce` in the code. The exact form uses the inverse Langevin function  $\mathcal{L}(x) = \coth x - 1/x$ : with  $\lambda_{ch} = \sqrt{I_1/3}$  and  $\beta = \mathcal{L}^{-1}(\lambda_{ch}/\sqrt{N})$ ,

$$\Psi = \mu N \left[ \frac{\lambda_{ch}}{\sqrt{N}} \beta + \ln \frac{\beta}{\sinh \beta} \right] + \text{const}, \quad \Psi_1 = \frac{\mu}{6} \frac{\sqrt{N}}{\lambda_{ch}} \mathcal{L}^{-1}\left(\frac{\lambda_{ch}}{\sqrt{N}}\right). \quad (26)$$

### 3.6 Ogden

Expressed directly in the principal stretches,

$$\Psi = \sum_{r=1}^M \frac{\mu_r}{\alpha_r} (\lambda_1^{\alpha_r} + \lambda_2^{\alpha_r} + \lambda_3^{\alpha_r} - 3), \quad \lambda_i \frac{\partial \Psi}{\partial \lambda_i} = \sum_r \mu_r \lambda_i^{\alpha_r}, \quad \mu = \frac{1}{2} \sum_r \mu_r \alpha_r, \quad (27)$$

with  $\mu_r \alpha_r > 0$  for each term. The exponents need not be integers. Neo-Hookean is  $M = 1$ ,  $\alpha_1 = 2$ ,  $\mu_1 = \mu$ ; Mooney–Rivlin is  $M = 2$ ,  $(\alpha_1, \alpha_2) = (2, -2)$ ,  $(\mu_1, \mu_2) = (2c_1, -2c_2)$ , using  $\sum \lambda_i^{-2} = I_2$ . Ogden’s classical three-term fit to Treloar’s vulcanised-rubber data is  $(\mu_r) = (0.63, 0.0012, -0.01)$  MPa,  $(\alpha_r) = (1.3, 5.0, -2.0)$ . Because it is not an invariant model, Ogden’s stresses are evaluated in the principal frame with (10)–(12). This is **ogden** in the code, which evaluates it spectrally (Jacobi eigen-decomposition of  $\mathbf{C}$ ) and forms the tangent analytically from the spectral representation, with the usual limit at coincident principal stretches, since eigenvectors are not differentiable there.

| Model         | $\Psi$                                                            | Parameters               | $\mu$                               |
|---------------|-------------------------------------------------------------------|--------------------------|-------------------------------------|
| Neo-Hookean   | $\frac{\mu}{2}(I_1 - 3)$                                          | $\mu$                    | $\mu$                               |
| Mooney–Rivlin | $c_1(I_1 - 3) + c_2(I_2 - 3)$                                     | $c_1, c_2$               | $2(c_1 + c_2)$                      |
| Yeoh          | $\sum_{i=1}^3 C_{i0}(I_1 - 3)^i$                                  | $C_{10}, C_{20}, C_{30}$ | $2C_{10}$                           |
| Gent          | $-\frac{\mu J_m}{2} \ln(1 - \frac{I_1 - 3}{J_m})$                 | $\mu, J_m$               | $\mu$                               |
| Arruda–Boyce  | $\mu \sum_{i=1}^5 C_i N^{1-i} (I_1^i - 3^i)$                      | $\mu, N$                 | $\mu(1 + \frac{3}{5N} + \dots)$     |
| Ogden         | $\sum_r \frac{\mu_r}{\alpha_r} (\sum_i \lambda_i^{\alpha_r} - 3)$ | $\mu_r, \alpha_r$        | $\frac{1}{2} \sum_r \mu_r \alpha_r$ |

Table 1: Incompressible isotropic models and their small-strain shear modulus.

**$I_1$ -only models.** Yeoh, Gent and Arruda–Boyce depend on  $I_1$  alone, so  $\Psi_2 = 0$  and (9) reduces to the neo-Hookean form with a deformation-dependent modulus,

$$\boldsymbol{\sigma} = \mu_{\text{eff}}(I_1) \text{dev } \mathbf{B} + p \mathbf{I}, \quad \mu_{\text{eff}}(I_1) = 2\Psi_1(I_1). \quad (28)$$

Every neo-Hookean result in Section 4 therefore applies to these models after replacing  $\mu$  by  $\mu_{\text{eff}}$  evaluated at the  $I_1$  of that deformation.

## 4 Homogeneous Deformations

A homogeneous deformation has constant  $\mathbf{F}$ , hence constant  $\boldsymbol{\sigma}$  and  $\mathbf{P}$ ;  $\text{Div } \mathbf{P} = 0$  holds identically, so the state is an exact solution of the equilibrium equations without body force, and  $p$  is a constant fixed by the traction-free faces. For each mode below the results are given for a general  $\Psi(I_1, I_2)$ , then specialised to neo-Hookean (NH), Mooney–Rivlin (MR) and Ogden;  $I_1$ -only models use (28). Extension is  $\lambda > 1$ , compression  $\lambda < 1$ ; the formulas cover both.

### 4.1 Uniaxial tension and compression

$$\mathbf{F} = \text{diag}(\lambda, \lambda^{-1/2}, \lambda^{-1/2}), \quad I_1 = \lambda^2 + \frac{2}{\lambda}, \quad I_2 = 2\lambda + \frac{1}{\lambda^2}, \quad \sigma_2 = \sigma_3 = 0. \quad (29)$$

From (12) with  $\lambda_1 = \lambda$ ,  $\lambda_2 = \lambda^{-1/2}$ ,

$$\sigma_1 = 2\left(\lambda^2 - \frac{1}{\lambda}\right)\left(\Psi_1 + \frac{\Psi_2}{\lambda}\right), \quad P_1 = \frac{\sigma_1}{\lambda} = 2\left(\lambda - \frac{1}{\lambda^2}\right)\left(\Psi_1 + \frac{\Psi_2}{\lambda}\right), \quad p = \frac{\sigma_1}{3}. \quad (30)$$

$$\text{NH: } \sigma_1 = \mu \left( \lambda^2 - \frac{1}{\lambda} \right), \quad P_1 = \mu \left( \lambda - \frac{1}{\lambda^2} \right), \quad (31)$$

$$\text{MR: } \sigma_1 = 2 \left( c_1 + \frac{c_2}{\lambda} \right) \left( \lambda^2 - \frac{1}{\lambda} \right), \quad P_1 = 2 \left( c_1 + \frac{c_2}{\lambda} \right) \left( \lambda - \frac{1}{\lambda^2} \right), \quad (32)$$

$$\text{Ogden: } \sigma_1 = \sum_r \mu_r (\lambda^{\alpha_r} - \lambda^{-\alpha_r/2}), \quad P_1 = \sum_r \mu_r (\lambda^{\alpha_r-1} - \lambda^{-\alpha_r/2-1}). \quad (33)$$

Small strain:  $\sigma_1 \approx 3\mu(\lambda - 1) = E(\lambda - 1)$ .

## 4.2 Equibiaxial tension

$$\mathbf{F} = \text{diag}(\lambda, \lambda, \lambda^{-2}), \quad I_1 = 2\lambda^2 + \frac{1}{\lambda^4}, \quad I_2 = \lambda^4 + \frac{2}{\lambda^2}, \quad \sigma_3 = 0. \quad (34)$$

$$\boxed{\sigma_1 = \sigma_2 = 2 \left( \lambda^2 - \frac{1}{\lambda^4} \right) (\Psi_1 + \lambda^2 \Psi_2), \quad P_1 = P_2 = \frac{\sigma_1}{\lambda}, \quad p = \frac{2\sigma_1}{3}.} \quad (35)$$

$$\text{NH: } \sigma_1 = \mu \left( \lambda^2 - \frac{1}{\lambda^4} \right), \quad (36)$$

$$\text{MR: } \sigma_1 = 2(c_1 + c_2 \lambda^2) \left( \lambda^2 - \frac{1}{\lambda^4} \right), \quad (37)$$

$$\text{Ogden: } \sigma_1 = \sum_r \mu_r (\lambda^{\alpha_r} - \lambda^{-2\alpha_r}). \quad (38)$$

Equibiaxial tension with stretch  $\lambda$  is uniaxial compression to  $\lambda^{-2}$  plus a hydrostatic stress, which is why the two tests are interchangeable for identifying  $\Psi$ .

## 4.3 Pure shear (planar tension)

$$\mathbf{F} = \text{diag}(\lambda, 1, \lambda^{-1}), \quad I_1 = I_2 = \lambda^2 + 1 + \frac{1}{\lambda^2}, \quad \sigma_3 = 0. \quad (39)$$

$$\boxed{\sigma_1 = 2 \left( \lambda^2 - \frac{1}{\lambda^2} \right) (\Psi_1 + \Psi_2), \quad \sigma_2 = 2 \left( 1 - \frac{1}{\lambda^2} \right) (\Psi_1 + \lambda^2 \Psi_2),} \quad (40)$$

$$P_1 = \frac{\sigma_1}{\lambda}, \quad P_2 = \sigma_2, \quad p = \frac{\sigma_1 + \sigma_2}{3}.$$

$$\text{NH: } \sigma_1 = \mu \left( \lambda^2 - \frac{1}{\lambda^2} \right), \quad \sigma_2 = \mu \left( 1 - \frac{1}{\lambda^2} \right), \quad (41)$$

$$\text{MR: } \sigma_1 = \mu \left( \lambda^2 - \frac{1}{\lambda^2} \right), \quad \mu = 2(c_1 + c_2), \quad \sigma_2 = 2(c_1 + c_2 \lambda^2) \left( 1 - \frac{1}{\lambda^2} \right), \quad (42)$$

$$\text{Ogden: } \sigma_1 = \sum_r \mu_r (\lambda^{\alpha_r} - \lambda^{-\alpha_r}), \quad \sigma_2 = \sum_r \mu_r (1 - \lambda^{-\alpha_r}). \quad (43)$$

The name refers to the principal stretches  $(\lambda, 1, \lambda^{-1})$ , which are those of a simple shear of amount  $\gamma = \lambda - \lambda^{-1}$ ; pure shear is simple shear followed by a rotation. Note that for Mooney–Rivlin  $\sigma_1$  depends only on  $\mu$ ; the constrained-direction stress  $\sigma_2$  is what distinguishes  $c_1$  from  $c_2$ .

#### 4.4 Simple shear

$$\mathbf{F} = \mathbf{I} + \gamma \mathbf{e}_1 \otimes \mathbf{e}_2 \quad (x_1 = X_1 + \gamma X_2), \quad \mathbf{B} = \begin{pmatrix} 1 + \gamma^2 & \gamma & 0 \\ \gamma & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{B}^{-1} = \begin{pmatrix} 1 & -\gamma & 0 \\ -\gamma & 1 + \gamma^2 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (44)$$

with  $I_1 = I_2 = 3 + \gamma^2$ . The principal stretches are  $\lambda = \frac{1}{2}(\gamma + \sqrt{\gamma^2 + 4})$ ,  $\lambda^{-1}$  and 1 (so  $\gamma = \lambda - \lambda^{-1}$ ), with the in-plane principal directions at an angle  $\theta$  to  $\mathbf{e}_1$  given by  $\tan 2\theta = 2/\gamma$ . The deviatoric stress (9) is

$$\sigma'_{12} = 2(\Psi_1 + \Psi_2)\gamma, \quad \sigma'_{11} = \frac{2}{3}(2\Psi_1 + \Psi_2)\gamma^2, \quad \sigma'_{22} = -\frac{2}{3}(\Psi_1 + 2\Psi_2)\gamma^2, \quad \sigma'_{33} = -\frac{2}{3}(\Psi_1 - \Psi_2)\gamma^2, \quad (45)$$

which gives the shear stress and the normal-stress differences independently of  $p$ :

$$\boxed{\begin{aligned} \sigma_{12} &= 2(\Psi_1 + \Psi_2)\gamma, & \sigma_{11} - \sigma_{22} &= \gamma\sigma_{12} = 2(\Psi_1 + \Psi_2)\gamma^2, \\ \sigma_{11} - \sigma_{33} &= 2\Psi_1\gamma^2, & \sigma_{22} - \sigma_{33} &= -2\Psi_2\gamma^2. \end{aligned}} \quad (46)$$

The first relation defines the secant shear modulus  $\mu(\gamma) = \sigma_{12}/\gamma = 2(\Psi_1 + \Psi_2)$ , which is constant for neo-Hookean and Mooney–Rivlin (both predict a linear shear response) and depends on  $\gamma$  through  $I_1 = 3 + \gamma^2$  for the  $I_1$ -only models. The relation  $\sigma_{11} - \sigma_{22} = \gamma\sigma_{12}$  is universal (independent of  $\Psi$ ). The mean stress depends on which face is traction-free:

$$\sigma_{33} = 0 \text{ (thin sheet): } \sigma_{11} = 2\Psi_1\gamma^2, \quad \sigma_{22} = -2\Psi_2\gamma^2, \quad p = \frac{2}{3}(\Psi_1 - \Psi_2)\gamma^2, \quad (47)$$

$$\sigma_{22} = 0 \text{ (free top face): } \sigma_{11} = 2(\Psi_1 + \Psi_2)\gamma^2, \quad \sigma_{33} = 2\Psi_2\gamma^2, \quad p = \frac{2}{3}(\Psi_1 + 2\Psi_2)\gamma^2. \quad (48)$$

The second case is the plane-strain one. Specialised:

$$\text{NH (either case): } \sigma_{12} = \mu\gamma, \quad \sigma_{11} = \mu\gamma^2, \quad \sigma_{22} = \sigma_{33} = 0, \quad p = \frac{\mu\gamma^2}{3}, \quad (49)$$

$$\text{MR, } \sigma_{33} = 0: \quad \sigma_{12} = \mu\gamma, \quad \sigma_{11} = 2c_1\gamma^2, \quad \sigma_{22} = -2c_2\gamma^2, \quad p = \frac{2(c_1 - c_2)\gamma^2}{3}, \quad (50)$$

$$\text{MR, } \sigma_{22} = 0: \quad \sigma_{12} = \mu\gamma, \quad \sigma_{11} = \mu\gamma^2, \quad \sigma_{33} = 2c_2\gamma^2, \quad p = \frac{2(c_1 + 2c_2)\gamma^2}{3}, \quad (51)$$

$$\begin{aligned} \text{Ogden: } \sigma_{12} &= \frac{\sum_r \mu_r (\lambda^{\alpha_r} - \lambda^{-\alpha_r})}{\lambda + \lambda^{-1}}, & \sigma_{11} - \sigma_{22} &= \gamma\sigma_{12}, \\ \sigma_{11} + \sigma_{22} - 2\sigma_{33} &= \sum_r \mu_r (\lambda^{\alpha_r} + \lambda^{-\alpha_r} - 2). \end{aligned} \quad (52)$$

The nominal stress follows from  $\mathbf{P} = \boldsymbol{\sigma} \mathbf{F}^{-T}$  with  $\mathbf{F}^{-T} = \mathbf{I} - \gamma \mathbf{e}_2 \otimes \mathbf{e}_1$ :

$$P_{11} = \sigma_{11} - \gamma\sigma_{12}, \quad P_{12} = \sigma_{12}, \quad P_{21} = \sigma_{12} - \gamma\sigma_{22}, \quad P_{22} = \sigma_{22}, \quad P_{33} = \sigma_{33}, \quad (53)$$

so that for neo-Hookean  $P_{11} = P_{22} = 0$  and  $P_{12} = P_{21} = \mu\gamma$ : the nominal tractions on the sheared faces are purely tangential. For any other model the lateral faces  $X_1 = \text{const}$  must carry the normal nominal traction  $P_{11}$  for the simple shear to be an exact solution.

#### 4.5 Consistency checks

All stresses vanish at  $\lambda = 1$  or  $\gamma = 0$ . The initial slopes, with  $\mu$  from (15), are

$$\left. \frac{d\sigma_1}{d\lambda} \right|_{\lambda=1} = \begin{cases} 3\mu & \text{uniaxial} \\ 6\mu & \text{equibiaxial} \\ 4\mu & \text{pure shear,} \end{cases} \quad \left. \frac{d\sigma_{12}}{d\gamma} \right|_{\gamma=0} = \mu, \quad (54)$$

and the normal-stress relation  $\sigma_{11} - \sigma_{22} = \gamma\sigma_{12}$  in simple shear holds for every isotropic incompressible model.

| Mode         | Stretches                                                     | $I_1$                          | Non-zero Cauchy stress                                                       | $p = \frac{1}{3} \text{tr } \boldsymbol{\sigma}$ |
|--------------|---------------------------------------------------------------|--------------------------------|------------------------------------------------------------------------------|--------------------------------------------------|
| Uniaxial     | $(\lambda, \lambda^{-1/2}, \lambda^{-1/2})$                   | $\lambda^2 + 2\lambda^{-1}$    | $\sigma_1 = \mu(\lambda^2 - \lambda^{-1})$                                   | $\sigma_1/3$                                     |
| Equibiaxial  | $(\lambda, \lambda, \lambda^{-2})$                            | $2\lambda^2 + \lambda^{-4}$    | $\sigma_1 = \sigma_2 = \mu(\lambda^2 - \lambda^{-4})$                        | $2\sigma_1/3$                                    |
| Pure shear   | $(\lambda, 1, \lambda^{-1})$                                  | $\lambda^2 + 1 + \lambda^{-2}$ | $\sigma_1 = \mu(\lambda^2 - \lambda^{-2}), \sigma_2 = \mu(1 - \lambda^{-2})$ | $(\sigma_1 + \sigma_2)/3$                        |
| Simple shear | $(\lambda, \lambda^{-1}, 1), \gamma = \lambda - \lambda^{-1}$ | $3 + \gamma^2$                 | $\sigma_{12} = \mu\gamma, \sigma_{11} = \mu\gamma^2$                         | $\mu\gamma^2/3$                                  |

Table 2: Neo-Hookean homogeneous solutions (one face traction-free as stated in the text). For Yeoh, Gent and Arruda–Boyce replace  $\mu$  by  $2\Psi_1(I_1)$  at the tabulated  $I_1$ .

## 5 Plane Strain and Reproducing These States in the Code

In two dimensions the kernels build  $\mathbf{F} = \begin{pmatrix} \mathbf{F}_{2D} & 0 \\ 0 & 1 \end{pmatrix}$  and  $J = \det \mathbf{F}_{2D}$ , i.e. plane strain with  $\lambda_3 = 1$  (the default, `plane: strain`; Section 5.3 for plane stress). An incompressible plane-strain state therefore has in-plane principal stretches  $(\lambda, \lambda^{-1})$ : up to a rotation every such homogeneous state is a pure shear (40) with  $\mathbf{e}_3$  as the constrained direction. The out-of-plane stress  $\sigma_{33}$  is a reaction and is generally non-zero; it enters both the pressure field  $p = \frac{1}{3}(\sigma_{11} + \sigma_{22} + \sigma_{33})$  and the `vonmises` output, which is computed from the full  $3 \times 3$  Cauchy stress,  $\sigma_{vm} = \sqrt{\frac{3}{2} \boldsymbol{\sigma}' : \boldsymbol{\sigma}'}$ . The code can also write  $\boldsymbol{\sigma}$  (six components),  $\mathbf{P}$  and  $\mathbf{F}$  (nine) and the energy density themselves. All such quantities are computed at the quadrature points and then presented as a continuous nodal field, as element averages, or as the raw point cloud (`output.fields`, `output.quadrature.at`, `output.nodal_projection`).

### 5.1 Plane-strain extension

Stretch  $\lambda$  along  $\mathbf{e}_1$  with the faces  $X_2 = 0, H$  traction-free ( $\sigma_{22} = 0$ ). Relabelling (40) with  $(\lambda_1, \lambda_2, \lambda_3) = (\lambda, \lambda^{-1}, 1)$ :

$$\mathbf{u} = (\lambda - 1)X_1 \mathbf{e}_1 + (\lambda^{-1} - 1)X_2 \mathbf{e}_2, \quad \sigma_{11} = 2(\Psi_1 + \Psi_2)\left(\lambda^2 - \frac{1}{\lambda^2}\right), \quad \sigma_{33} = 2\left(1 - \frac{1}{\lambda^2}\right)(\Psi_1 + \lambda^2\Psi_2), \quad (55)$$

$$p = \frac{\sigma_{11} + \sigma_{33}}{3}, \quad P_{11} = \frac{\sigma_{11}}{\lambda}, \quad \sigma_{vm} = \sqrt{\sigma_{11}^2 + \sigma_{33}^2 - \sigma_{11}\sigma_{33}}. \quad (56)$$

$$\text{NH: } \sigma_{11} = \mu\left(\lambda^2 - \frac{1}{\lambda^2}\right), \quad \sigma_{33} = \mu\left(1 - \frac{1}{\lambda^2}\right), \quad p = \frac{\mu}{3}\left(\lambda^2 + 1 - \frac{2}{\lambda^2}\right), \quad (57)$$

$$\text{MR: } \sigma_{11} = \mu\left(\lambda^2 - \frac{1}{\lambda^2}\right), \quad \sigma_{33} = 2(c_1 + c_2\lambda^2)\left(1 - \frac{1}{\lambda^2}\right), \quad p = \frac{\sigma_{11} + \sigma_{33}}{3}. \quad (58)$$

To realise this in the code (Dirichlet conditions are applied to the full displacement vector), prescribe the affine  $\mathbf{u}$  above on the two end faces  $X_1 = 0$  and  $X_1 = L$  and leave  $X_2 = 0, H$  free; alternatively prescribe  $\mathbf{u}$  on  $X_1 = 0$  and the nominal traction  $\mathbf{t}_R = P_{11} \mathbf{e}_1$  on  $X_1 = L$ . With Taylor–Hood spaces the affine displacement and the constant pressure lie in the discrete spaces, so the mixed



formulation reproduces  $\mathbf{u}$ , the constant  $p$  above and  $\sigma_{vm}$  to solver tolerance. The example meshes are generated by Gmsh from `apps/mesh/*.geo` with named physical groups (`left`, `right`, `top`, `bottom`), and the input files name the faces rather than numbering them.

## 5.2 Plane-strain simple shear

$\mathbf{u} = \gamma X_2 \mathbf{e}_1$ . With the displacement prescribed on the whole boundary (the setting of the patch test in `tests/test_mixed.cpp`) the incompressible pressure is determined only up to a constant, because adding a constant to  $p$  adds  $c \int J \mathbf{F}^{-T} : \text{Grad } \mathbf{w} \, dV = c \int_{\partial \Omega_R} \mathbf{w} \cdot J \mathbf{F}^{-T} \mathbf{N} \, dA = 0$  to the residual for every admissible  $\mathbf{w}$  (Piola identity). The checkable quantities are then the deviatoric stress  $\boldsymbol{\sigma}'$ , the shear stress  $\sigma_{12} = 2(\Psi_1 + \Psi_2)\gamma$  and the normal-stress differences of Section 4; at finite  $\kappa$  the affine field is exactly isochoric, so  $p = \kappa(J - 1) = 0$  and  $\boldsymbol{\sigma} = \boldsymbol{\sigma}'$ . If instead only the top and bottom faces are constrained and the lateral faces are left free, simple shear is *not* the exact solution (the lateral faces would have to carry  $P_{11} = \sigma_{11} - \gamma\sigma_{12}$  and  $P_{21} = \sigma_{12} - \gamma\sigma_{22}$ ); the neo-Hookean model is the exception, with  $P_{11} = 0$  but  $P_{21} = \mu\gamma \neq 0$ , so even there free lateral faces perturb the state near the edges.

## 5.3 Plane stress

For a thin sheet loaded in its plane the appropriate two-dimensional idealisation is plane stress,  $\sigma_{i3} = 0$ . The thickness stretch  $\lambda_3 = F_{33}$  is then not prescribed but set by the material. For an incompressible material  $\lambda_3 = 1/\det \mathbf{F}_{2D}$ , and  $\sigma_{33} = 0$  fixes the mean stress as  $p = -\sigma'_{33}$ , so

$$\sigma_{\alpha\beta} = \sigma'_{\alpha\beta} - \sigma'_{33} \delta_{\alpha\beta}, \quad \mathbf{P} = \mathbf{P}_{\text{iso}} + p \mathbf{F}^{-T}, \quad p = -\lambda_3 P_{\text{iso},33}. \quad (59)$$

The constraint and its multiplier are eliminated pointwise: the sheet problem is a displacement problem in the two in-plane components, with no pressure unknown, no inf-sup condition and no locking. For a compressible material  $\lambda_3$  solves  $P_{33}(\mathbf{F}_{2D}, \lambda_3) = 0$  at every point. The states of Section 4 are all plane-stress states, so they become two-dimensional problems: the uniaxial sheet with  $\mathbf{u} = (\lambda - 1)X_1 \mathbf{e}_1 + (\lambda^{-1/2} - 1)X_2 \mathbf{e}_2$  on the end faces and the lateral faces free, and equibiaxial tension, pure shear and thin-sheet simple shear ( $\sigma_{33} = 0$ , Section 4.4) with the affine displacement on the whole boundary, which is well posed here because  $p$  is determined pointwise (contrast Section 5.2). The code selects it with `plane: stress` (2D, displacement formulation); the material is wrapped in an adapter that returns  $\mathbf{P}$  at the thickness stretch, so the tangent obtained by dual numbers is the consistent plane-stress tangent (for a compressible material the converged root is differentiated implicitly). The thickness stretch is the output field `thickness_stretch`, and `jacobian` is the true  $J$  (one for an incompressible sheet). Since each load step starts with the boundary displaced and the interior lagging, shear-dominated sheet problems need increments that shear a boundary layer of elements by no more than a few percent, as the energy grows like  $(\det \mathbf{F}_{2D})^{-2}$ .

## 5.4 Parameter correspondence

The `pressure` output field is  $p = \frac{1}{3} \text{tr } \boldsymbol{\sigma}$ , tensile positive; the first Piola–Kirchhoff stress assembled by the mixed kernel is  $\mathbf{P} = \mathbf{P}_{\text{iso}} + p J \mathbf{F}^{-T}$ . A compressive hydrostatic state therefore has  $p < 0$ .

## 5.5 Verification inputs

The inputs `apps/input/finite_elasticity/homogeneous/plane_strain-<model>.yaml` (unit square `apps/mesh/square.msh`, plane-strain extension to  $\lambda = 2$  as in Section 5.1),

| Model               | material keys                                                                         | Notes                                                 |
|---------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------|
| Neo-Hookean         | <code>model: iso_neo_hookean, mu</code>                                               | or $E$ , $\nu$ with<br>$\mu = E/(2(1 + \nu))$         |
| Mooney–Rivlin       | <code>model: mooney_rivlin, c1, c2</code>                                             | $\mu = 2(c_1 + c_2)$                                  |
| Yeoh                | <code>model: yeoh, c10, c20, c30</code>                                               | $\mu = 2C_{10}$ ; $c_{20}$ , $c_{30}$ default<br>to 0 |
| Gent                | <code>model: gent, mu, Jm</code>                                                      |                                                       |
| Arruda–Boyce        | <code>model: arruda_boyce, mu, N</code>                                               | $\mu$ is $nkT$ , $\mu_0$ from (25)                    |
| Ogden               | <code>model: ogden, mu_r, alpha_r</code>                                              | lists of up to six terms                              |
| Incompressible      | <code>incompressible: true</code> or <code>nu: 0.5</code>                             | needs <code>formulation: mixed</code>                 |
| Near-incompressible | <code>kappa: <math>\kappa</math></code> or <code>nu: <math>\nu &lt; 0.5</math></code> | $p = \kappa(J - 1)$ , either<br>formulation           |

Table 3: Mapping between the models above and the material block of the input file.

`plane_stress_<model>.yaml` (the same sheet in plane stress: uniaxial tension to  $\lambda = 2$ , thickness stretch  $\lambda^{-1/2}$ , no pressure unknown) and `uniaxial_<model>.yaml` (unit cube `apps/mesh/cube.msh`, uniaxial tension to  $\lambda = 2$ , the affine displacement of Section 4.1 prescribed on the faces `left` and `right`, the four lateral faces free) exist for all six models, together with equibiaxial and pure-shear cubes (`equibiaxial_<model>.yaml`, `pure_shear_<model>.yaml`; Sections 4.2 and 4.3, the faces  $Z = 0, 1$  free) and the uniaxial symmetry model with rollers. Their Dirichlet entries write the affine data  $\mathbf{u} = \mathbf{G}\mathbf{X}$ ,  $\mathbf{G} = \text{diag}(\lambda_a - 1)$ , as expression strings (e.g. `["x", "-0.5*y"]`). `apps/homogeneous_compare.py` runs them and prints the probed displacement, pressure, von Mises stress,  $J$ , and the components of  $\boldsymbol{\sigma}$  and  $\mathbf{P}$  next to the closed forms of Sections 4.1 and 5.1 (`make homogeneous`); all agree to about  $10^{-13}$  with quadratic Newton convergence, in the nodal and the element presentations alike. `tests/test_homogeneous` (part of `make check`) repeats the comparison programmatically: at the material point, the code's Cauchy stress with the analytic mean stress against the principal stresses of Sections 4.1–4.3 and 5.1 and the deviatoric stress of simple shear (Section 4.4); as finite element solutions of the mixed formulation in 2D and 3D, with every quadrature quantity checked in its nodal (both projections), element-average and quadrature-point presentations; and as plane-stress solutions of the sheet states uniaxial, equibiaxial, pure shear and simple shear (Section 5.3), including the Mooney–Rivlin  $\sigma_{22} = -2c_2\gamma^2$  of simple shear.

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